KKK-L-1542C
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SUPERSEDING
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FEDERAL SPECIFICATION

LOADERS, SCOOP TYPE, FOUR-WHEEL DRIVE, PNEUMATIC-TIRED

This specification was approved by the General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers rear-engine mounted, power shift, four-wheel drive, pneumatic-tired, hydraulically operated, scoop type, front-end loaders.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification or which are commercially available but are intended to cover only those used by the Federal Government.

1.2 Classification.

1.2.1 Sizes. The four-wheel drive loaders covered by this specification shall be furnished in the sizes listed in tables I, II, and III, as specified (see 6.1).

FSC 3805

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TABLE I. LOADER, MINIMUM CAPACITY, WEIGHT, HORSEPOWER, DRIVE, AND TIRE REQUIREMENTS

4-WHEEL DR. LOADER SIZE	BUCKET SAE RATED CAPACITY IN CU. YDS. OF 3,000 LB. MATERIAL 1/ GP*	MP* OR ROCK	MINIMUM TIPPING LOAD, AT FULL TURN LBS. 2/ GP*	MP* OR ROCK	MINIMUM WEIGHT WITH BUCKET LBS. 3/ GP*	MP* OR ROCK	MIN. APPL. N.H.P. 4/	MIN. NO. OF FWD. AND REV. SPD. RNGS.	MAX. CLEAR. TURN. RAD. (FT.) W/O BCKT. 5/	MIN. TREAD WIDTH INS. CTR. TO CTR. 6/	MINIMUM TUBELESS TIRE SIZE AND PLY RATING (OR TIRES W/NO LESS LOAD RATING AT 5 M.P.H.) 7/	
LW-1 YD	1 YD	3/4 YD	6000	4500	10800	11400	50	3-1	17.5	60	15.0x25	8
LW-1 1/4 YD	1 1/4 YD	1 YD	7500	6000	11200	11600	65	3-2	17.5	65	15.5x25	8
LW-1 1/2 YD	1 1/2 YD	1 1/4 YD	9000	7500	13700	14700	80	3-2	17.6	65	15.5x25	8
LW-2 YD	2 YD	1 3/4 YD	12000	10500	17700	18500	100	3-2	18.6	68	17.5x25	12
LW-2 1/2 YD	2 1/2 YD	2 YD	15000	12000	22000	23000	120	3-2	20.0	72	20.5x25	12
LW-2 3/4 YD	2 3/4 YD	2 1/4 YD	16500	13500	25000	26000	130	3-2	20.0	73	20.5x25	12
LW-3 YD	3 YD	2 1/2 YD	18000	15000	27000	28000	150	3-2	22.0	74	20.5x25	12
LW-3 1/2 YD	3 1/2 YD	3 YD	21000	18000	30000	31000	170	3-2	22.0	77	23.5x25	12
LW-4 YD	4 YD	3 1/2 YD	24000	21000	35000	36000	185	3-2	22.6	82	23.5x25	12
LW-4 1/2 YD	4 1/2 YD	4 YD	27000	24000	40000	41000	230	3-3	22.6	87	26.5x25	20
LW-5 YD	5 YD	5 YD	30000	30000	49000	50000	260	3-3	23.6	89	26.5x25	20
LW-5 1/2 YD	5 1/2 YD	5 1/2 YD	33000	33000	50000	51000	270	3-3	24.6	93	26.5x25	20
LW-6 YD	6 YD	6 YD	36000	36000	57000	58000	285	3-3	26.6	95	29.5x25	22
LW-7 YD	7 YD	7 YD	42000	42000	79500	80000	360	3-3	26.6	101	29.5x25	22

*EQUALS GENERAL PURPOSE OR MULTI-PURPOSE

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NOTES: SEE TABLE I

1/ Minimum heaped capacity of bucket shall be rated in accordance with SAE J742. Loader must have adequate lifting capacity at pressures below manufacturer's standard relief valve setting to raise the specified size and type bucket from ground level to maximum hinge pin height loaded to rated capacity with material weighing 3,000 lbs. per cu. yd. If a bucket capacity other than that listed in Table I is specified, it will be assumed that: (a) service requirements dictate a different material weight than 3,000 pounds per cubic yard and (b) the loader when equipped with this bucket meets the tipping and operating load requirements for the loader size specified. LW size 5 yards thru 7 yards are normally rock buckets.

2/ Minimum tipping load in pounds shall be measured in accordance with SAE J732C. Articulated loaders must be in full turn position. Operating load shall be as recommended in SAE J818. Tipping load shall be measured without hydroinflation in tires.

3/ Loader minimum shipping weight with specified bucket shall be complete with hydraulic controls, standard accessories and attachments, specified tires, and counterweight when furnished with loader and ROPS unless otherwise specified. Multipurpose bucket shall be designed for use as a clamshell, scraper and dozer, as well as front-end shovel, all hydraulically controlled from operator's seat.

4/ Applies net horsepower at flywheel of fully equipped engine at rated governed r.p.m. as installed in loader, including power deductions for fan, generator or alternator, and waterpump. Manufacturers shall furnish certified net horsepower and torque curves corrected to SAE J816 standard conditions of 500 feet altitude (29.38 inches of mercury barometric pressure minus 0.38-inch mercury humidity vapor pressure) and ambient air temperature of 85°F. for nonturbocharged engines; however, factory observed horsepower and torque curves, certified for 500 feet or more altitude, and 85°F. for turbocharged engines, shall be furnished (see 3.7.1 and 4.3).

5/ Maximum clearance turning radius shall be measured at outside of wheels or chassis without bucket, whichever is greater. Turning radius with bucket is not specified because of variations in types of buckets and whether with or without teeth.

6/ Minimum tread width shall be of the narrowest if that of front and rear wheels are different.

7/ Tires on all wheels shall be the same tubeless and of the same type, size, and ply rating. Tire size and ply rating supplied shall have a load carrying capacity equal to or greater than the tire specified in Table I. Load carrying capacity will be as shown in current tire and rim association yearbook. Off-the-road slow speed service - 5 mph table. Under no circumstances is the tire load carrying capacity to be determined at a cold inflation pressure greater than that shown as maximum in the Tire and Rim tables for a particular ply rating. Tires shall be of tubeless type and shall have a tread pattern suitable for loader operation or as specified in the invitation for bid.

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TABLE II. LOADER LIFT AND BUCKET MECHANISM MINIMUM REQUIREMENTS

4-WHEEL DR. LOADER SIZE	MIN. HGT. TO CTR. OF HINGE PIN FULLY RAISED	BUCKET CUTTING EDGE CLEAR. FROM GROUND WHEN FRONT DUMPED AT 45° ANGLE		BUCKET MIN. CLEAR. FROM GROUND WHEN BOTTOM DUMPED	MAX. TIME IN SECONDS		GENERAL PURPOSE BUCKET MIN. REACH FROM TIRES AT 7 FT. 45° ANGLE DUMP	MIN. REACH FROM FRONT OF TRACTOR OR TIRES AT NOT LESS THAN BUCKET CUTTING EDGE HEIGHT WHEN FRONT DUMPED AT 45° ANGLE		DIGGING DEPTH		MIN. BUCKET BREAK-OUT AND TILT-BACK FROM HORIZONTAL (HINGE PIN RISE NOT OVER 18 INCHES)	MINIMUM LOADER BUCKET WIDTH
		GP*	MP OR ROCK	MP	RAISE BUCKET TO MIN. HINGE PIN HEIGHT	POWER LOWER BUCKET FROM MIN. HINGE PIN HEIGHT TO GROUND		GP*	MP* OR ROCK	DIGGING DEPTH WITH BUCKET CUTTING EDGE HORZ. POSITION GP*	DIGGING DEPTH WITH BUCKET CUTTING EDGE DOZER POSITION MP		
	INS.	INS.	INS.	INS.			INS.	INS.	INS.	INS.	INS.	DEGREES	
LW-1 YD	122	96	92	115	6.5	4.3	35	24	26	1	2	33	
LW-1 1/4 YD	122	96	99	120	6.5	4.5	41	28	30	1	2	40	
LW-1 1/2 YD	130	103	99	121	7.0	5.0	43	28	30	1	3	40	
LW-2 YD	130	103	102	125	7.0	5.3	50	30	31	1	4	40	
LW-2 1/2 YD	140	105	102	135	7.5	5.5	47	29	31	1	4	38	
LW-2 3/4 YD	140	106	103	137	7.5	5.5	48	31	33	1	4	38	
LW-3 YD	140	106	103	132	7.5	6.0	51	33	31	1	5	38	CUTTING WIDTH SHALL NOT BE LESS THAN MAXIMUM OVERALL WIDTH
LW-3 1/2 YD	150	110	107	137	7.5	5.7	58	33	38	1	5	40	
LW-4 YD	156	115	109	146	7.5	6.0	58	35	38	1	6	40	
LW-4 1/2 YD	158	115	110	--	8.0	6.0	64	40	44	1	6	40	
LW-5 YD	162	120	116	--	8.5	7.0	70	42	46	1	-	40	
LW-5 1/2 YD	164	121	118	--	8.5	7.0	70	44	50	1	-	40	
LW-6 YD	166	121	120	--	9.5	7.2	75	48	51	1	-	40	
LW-7 YD	189	135	137	--	9.5	7.3	87	52	52	1	-	39	

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TABLE III. MINIMUM TRAVEL SPEED IN M.P.H. AT RATED ENGINE R.P.M. INTO TRANSMISSION

LOADER SIZE	LW 1	LW 1 1/4	LW 1 1/2	LW 2	LW 2 1/2, 2 3/4	LW 3	LW 3 1/2	LW 4	LW 4 1/2	LW 5	LW 5 1/2	LW 6	LW 7
FORWARD:													
1ST M.P.H.	2.7	2.7	2.7	2.3	1.5	1.5	2.4	2.4	2.0	2.0	2.0	2.0	2.0
HIGH M.P.H.	15.0	14.5	15.0	18.5	18.0	18.0	17.5	17.5	18.0	18.0	18.0	18.0	18.0
REVERSE:													
1ST M.P.H.	3.0	3.0	3.0	2.0	2.0	2.0	3.0	3.0	2.0	2.0	2.0	2.0	2.0
HIGH M.P.H.	6.5	6.5	7.4	7.0	7.0	7.0	7.0	7.0	7.0	14.5	15.0	14.2	14.0

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2. APPLICABLE DOCUMENTS

2.1 Specification and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

O-E-915 - Extinguishers, Fire, Dry-Chemical (Portable).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration's Regional Offices in Boston; New York; Washington, D.C.; Atlanta; Chicago; Kansas City, MO; Fort Worth; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, Handbooks, and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-T-3351 - Tractor, Full Tracked Low Speed, Tractor, Wheeled, Agriculture and Tractor, Wheeled, Industrial, and Their Attachments, Packaging of.

Military Standard:

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

Laws and Regulations

29 CFR 1926.600:

Equipment

29 CFR 1926.602:

Material handling equipment

29 CFR 1926.604:

Overhead covering

29 CFR 1926.1000:

Rollover protective structure (ROPS) for material handling equipment:

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Incorporated. Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tarriff Order Section, 1616 P Street, NW, Washington, D.C. 20036.)

Uniform Classification Committee Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Tire and Rim Association, Inc., Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Inc., Command Building, 34 N. Hawkins Avenue, Akron, Ohio 44313.)

Society of Automotive Engineers (SAE), Inc., Standards:

J53 - Minimum Performance Criteria for Emergency Steering of Wheeled Earth-moving Construction Machines.
J386 - Seatbelts.
J534 - Lubrication Fittings.
J732 - Specification Definitions - Industrial Tractor Front-End Loader.
J742 - Front End Loader Bucket Rating.
J816 - Engine Test Code - Nonturbocharged Spark Ignition and Diesel.
J818 - Operating Load for Front End Loaders.
J1152 - Minimum Performance Criteria for Brake System for Off-Highway, Rubber Tired Front End Loaders and Dozers.

Recommended Practices:

J350 - Spark Arrester Test Procedure for Medium Size Engines.
J394 - Minimum Performance Criteria for Roll-over Protective Structure for Rubber-Tired Front End Loaders and Rubber-Tired Dozers.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

Forest Service Standard:

5100-1a - Spark Arrester for Internal Combustion Engine.

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 General. This specification covers rear-engine mounted, four-wheel drive, pneumatic-tired, hydraulically operated, scoop-type, front-end loaders of integral heavy-duty construction, complete with all necessary operating accessories customarily furnished with power-shift loaders of this type, whether stipulated herein or not, together with such modifications and optional attachments as may be specified. The basic requirements are stability, dependability, and ample power to load, lift, carry, and dump rated capacity in the specified size and type bucket, together with mobility for fast time cycles and rapid self-transport over uneven terrain. Multipurpose bucket may be specified for more versatility (see 3.14.1).

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3.2 Material. The loaders and all parts thereof shall be made of materials which are suitable for the intended service and shall be produced by use of current standard manufacturing processes. Components, particularly of the electrical, fuel, air, and exhaust systems, shall resist harmful effects of dust, rust corrosion, wear, and moisture.

3.2.1 Recovered materials. All equipment, material, and articles incorporated in this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this document.

3.3 Standard products. The loader shall be essentially the standard current product of the manufacturer, differing therefrom only in respects necessary to meet special requirements (see 3.25). The component parts of the unit need not be the product of the same manufacturer.

3.4 Loader capacity and operational requirement. The loader shall have the specified bucket SAE rated capacity for material weighing 3,000 lbs. per cu. yd. and shall be capable of starting and operating efficiently under any of the conditions or combinations of conditions outlined in 3.4.1 and 3.4.2. When specified (3.24.1) a pin or clevis type drawbar shall be provided, except when a rear mounted backhoe, ripper, scarifier, or winch is specified which will not permit mounting of drawbar.

3.4.1 Temperature conditions. The vehicle shall be capable of starting and of satisfactory performance in any ambient temperature of -10°F. up to and including 110°F. with full impact of solar radiation of 360 B.t.u./sq. ft./hr. for at least 4 hours. When specified (see 6.1), it shall be capable of starting and operating at -20°F., -40°F., or -65°F. The low temperature requirement shall be met after the specified low temperature has been sustained for not less than 3 days without benefit of solar radiation, when serviced in accordance with the manufacturer's recommendation. The vehicle shall not be damaged by storage at any ambient temperature from plus 160°F. for periods of 4 hours per day to minus 40°F. (minus 65°F. when that performance temperature is specified) for periods of at least 3 days duration.

3.4.2 Use conditions. Design and construction shall be such that the loaders will withstand the extremely hard usage encountered in service such as digging, lifting, transporting, dumping materials, operation over rough terrain, and storage and operation in the open air under all weather conditions for extended periods of time.

3.4.3 Loader weight. For satisfactory operating load stability, the loader shall weigh not less than the applicable minimum given in table I. The loader weight shall be the shipping weight with specified bucket complete with hydraulic controls, starting and lighting system, standard accessories and attachments, specified tires, counterweight when furnished with loader, and ROPS unless otherwise specified. It shall include full complement of fuel, water, and lubricant excluding operator. Tire hydroinflation is not acceptable for meeting minimum weights or tipping in table I (see 6.2).

3.5 Ease of maintenance. The design of the loader shall permit ready removal and replacement of component units, adjustment, and removal of parts and accessories with minimum disturbance of other elements. The loader shall be equipped with high pressure fittings at all points where intermittent lubrication is required. Lubrication means shall be provided for all moving parts requiring lubrication. Hydraulic lubrication fittings shall be in accordance with Specification SAE J534. Pressure lubrication fittings shall not be used where normal lubricating pressure may damage seals or other parts.

3.6 Frame. Loader frames shall be fabricated steel reinforced and designed to withstand the maximum stresses normally imposed in heavy-duty loader operation. Articulated frames shall provide for rear axle oscillation in each direction from horizontal of not less than 10-1/2°. Front and rear articulated frames shall be joined by a rugged hinge assembly that allows free movement to either side but which holds the frames rigid against twisting forces. However, the articulation joint may be designed so that in addition to turning, it also allows oscillation of at least 10° between the front and rear axles, thus eliminating the need for a separate oscillation of the rear axle only.

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3.7 Diesel engine. The engine shall be of the compression ignition type, with two-or four-stroke cycle, capable of operating on commercial diesel fuel as recommended by the manufacturer. The engine shall be equipped with an adequate and fuel efficient injection mechanism, adequate fuel oil filter system, efficient lubricating oil filter, and a two stage dry type air cleaner with service indicator shall be furnished.

3.7.1 Engine horsepower. The loader shall have a rear-mounted engine that applies not less than the minimum horsepower at the engine flywheel shown in table I for the specified size. The applied net horsepower at the flywheel shall be that developed at the manufacturer's advertised rated governed speed of the engine as furnished with the loader with all standard accessories including cooling fan, water pump (if liquid cooled), and electrical equipment. Manufacturers shall furnish certified engine net horsepower and torque curves, corrected to SAE J816 standard test conditions for nonturbocharged engines, and factory observed curves certified for 500 feet or more altitude and 85°F for turbocharged engines.

3.7.2 Engine accessories.

3.7.2.1 Governor. The engine governor shall be of the centrifugal type. Governed speed setting throughout the engine load range shall be by foot-accelerator control.

3.7.2.2 Engine starting system. The starting system shall be of ample power to crank the engine at sufficient speed for starting when the engine is cold, down to an ambient temperature, as specified (see 3.4.1). Glow plugs or ether starting aid shall be furnished with diesel engines that have a direct electric starting system to permit starting at temperatures down to -10°F as required in paragraph 3.4.1. Starter shall not actuate unless transmission is in neutral, or a means shall be provided to automatically neutralize the transmission when engine is stopped and to permit restarting the engine without vehicle movement. When specified, an engine block heater with minimum 2 watts per cubic inch of engine displacement shall be provided (see 3.24.1), if liquid cooled is furnished.

3.7.2.3 Alternator. The alternator with integral regulator shall be the manufacturer's standard, adequate for use with the battery specified in 3.7.2.4 and with the electrical loads associated with paragraphs 3.16, 3.17, 3.18 and any other accessories consuming electrical current. Alternators with integral regulators will not show a discharge rate with all electrical components in the on position when engine is at idle.

3.7.2.4 Battery. The battery or batteries shall be the manufacturer's standard of the type regularly furnished for cold starting, as specified (see 3.4.1). Mounting shall permit ready accessibility for servicing, removing, and replacement.

3.7.2.5 Engine Cooling. The cooling system for a liquid-cooled engine shall maintain coolant at a temperature below the boiling point when working at rated rpm and operated at 10,000 feet above sea level or in an ambient air temperature of not less than 110°F. The radiator shall be protected by a heavy-duty guard or bumper plate and be provided with a shroud or radial flow shroud with blower fan. When the loader is equipped with a direct air cooling system, is working at rated rpm and operated at 10,000 feet above sea level, or in an ambient air temperature of not less than 110°F., the engine lubricating oil shall be maintained at a temperature below oil breakdown temperature.

3.7.2.6 Engine lubricating system. The manufacturer's current standard production engine lubricating system shall be acceptable.

3.7.2.7 Engine exhaust system. An exhaust pipe with a rain cap on a vertical exhaust, or a muffler with internal baffling and drain holes to prevent rain water intrusion, or the manufacturer's standard curved piping as regular equipment on a horizontal exhaust, shall be furnished. All nonturbocharged engines shall include an exhaust muffler. When specified (see 3.24.1), nonturbocharged diesel engines shall be equipped with a spark arrester having an arresting efficiency of 80 percent as determined by SAE J350 and, when specified, 5100-1a Forest Service standard arresting efficiency.

3.7.2.8 Hood side panels. Easy engine access hood side panels, doors, or shields shall be furnished.

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3.7.2.9 Crankcase protection. When specified (see 3.24.1), the loader shall be equipped with a crankcase guard capable of affording positive protection to the under parts of the engine and torque converter, unless the loader construction regularly provides this protection.

3.8 Fuel tank. The fuel tank shall have capacity for at least 8 hours of normal loader operation. The tank shall be equipped with a means for draining water and sediment from bottom of tank.

3.9 Engine power transmission. The loader shall have a torque converter/power shift transmission system, or hydrostatic drive system to transmit engine power to the driving wheels. The torque converted/power shift transmission shall provide not less than the number of forward and reverse speed ranges specified in Table I and at least the minimum traveling m.p.h. forward and reverse specified in Table III, or a hydrostatic drive system that provides infinite speed ranges up to the minimum travel speeds specified in Table III.

3.9.1 Torque converter. The hydraulic torque converter shall have the loader manufacturer's standard stall torque ratio. It shall be matched to the engine so that simultaneous crowding, loading, and lifting of the bucket can be done without excessive lug down in engine rated governed rpm. A twin-turbine torque converter, with automatic dual range drive, shall be accepted as doubling the number of transmission forward and reverse speed ranges. An efficient torque converter oil cooler and oil filtering system shall be supplied.

3.9.2 Transmission. The power-shift transmission shall be hydraulically actuated friction-clutch controlled countershaft or planetary type for effortless forward-reverse and speed range selection. First and high gear forward and reverse travel speeds in mph, at rated engine rpm into the transmission, shall be not less than specified in table III. A manual shift between two working ranges and two higher traveling range power shift forward and reverse speeds is acceptable as doubling the number of transmission speeds.

3.9.3 Hydrostatic drive system. The hydrostatic drive system shall consist of a hydraulic pump and motor combination to provide torque multiplication and power transmission from the engine to the driving wheels. The hydrostatic drive system shall provide automatic speed variation depending upon load demand. Top speeds, forward/reverse, shall be as stated in Table III. A manual shift between working and traveling speeds shall be provided. Forward and reverse directional control shall be manually operated.

3.10 Power train. The power train shall consist of drive shafts, differentials, and axles with planetary or gear reduction drives for all four wheels. All components shall be furnished with torque capacity exceeding the maximum to which they will be subjected at torque converter or hydrostatic drive stall in low gear with full power input from the engine, and shall be free of vibration and abnormal noise in all speeds and axle operating positions. When specified (see 3.24.1), power transfering differential shall be furnished in front or rear axle (not normally required on loaders LW 2-1/2 and larger).

3.11 Brakes. The loader service brake system, emergency stopping system and parking system shall conform to SAE 1152.

3.12 Tires. Tires on all wheels shall be of the same tubeless type and have a size and ply rating or equivalent load rating at 5 mph not less than specified in table I. Tread pattern shall be suitable for general loader operation. However, tires with increased ply rating or a specific tread type may be specified in the IFB to meet special application requirements (see 3.24.1). For satisfactory sidewise loader stability, the minimum wheel tread width between the center shall be not less than that given in table I.

3.13 Steering. Articulated steering shall be furnished. Articulated frame loaders shall be hinged between the front and rear axles to swing through a minimum angle of 35 degrees to the right or left of center actuated by hydraulic cylinders. A positive ratio between the amount the steering wheel is turned and the angle that the loader turns shall be maintained. Clearance turning radius without bucket shall not exceed that given in table I. When specified, emergency steering shall be provided conforming to SAE J53 in the event of engine or steering power source failure (see 3.24.1).

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3.14 Loader lift and bucket mechanism.

3.14.1 Bucket. Heavy-duty general purpose bucket of all-welded construction, with bolted or welded on replaceable wear and abrasion resistant cutting edges without teeth, shall be furnished, unless multipurpose type bucket with teeth is specified (see 6.1 and 3.24.1). Loaders equipped with rock buckets (see 3.24.1) shall weigh not less than specified under the multipurpose bucket minimum weight column in table I. Bucket cutting edge widths shall be not less than maximum overall loader width without bucket. When specified (see 3.24.1) teeth (replaceable type) for the general purpose bucket shall be furnished.

3.14.2 Loader mechanism requirements. The loader lift and bucket mechanism for each size loader and type of buckets shall meet the minimum requirements specified in table II without malfunction, breakage, or permanent deformation. For maximum operator safety, booms, bucket operating mechanism, and hydraulic cylinders shall be located ahead of the operator, or positive safety guards shall be provided.

3.14.3 Loader hydraulic system. The lift and bucket mechanism shall be actuated by double-acting hydraulic cylinders with chrome-plated piston rods. The hydraulic pump(s) shall be on a live drive from the engine and shall provide ample gpm to meet the minimum lift and power lower time requirements specified in table II and, also adequate pressure for actuating the lifting of the fully loaded bucket in all positions. Ample hydraulic circuit and reservoir capacity shall be furnished for sustained fast operating cycles without overheating in 110°F. ambient air temperature. An efficient oil filtering system shall be provided with filter element(s) readily accessible for removal and replacement. Hydraulic oil level must be easily checked and maintained.

3.15 Operator's seat and controls.

3.15.1 Seat. The operator's seat shall be front-located ahead of the engine within easy and convenient reach of all controls. It shall provide adjustments for comfortable riding positions and be upholstered with weather-resistant material suitable for severe service. The seat shall allow the operator a full view of the bucket in all positions, as well as good visibility of the work area. Handholds and steps or ladder shall be provided for easy and safe access to operator's compartment. Unless otherwise specified, well-anchored seatbelts shall be furnished conforming to SAE J386 (Type I). When specified (see 3.24.1), armrests shall be provided on loaders not equipped with bucket seats.

3.15.2 Loader hydraulic controls. Loader hydraulic lift control shall provide raise, hold, lower, and float positions. General purpose bucket control shall have tilt-back, hold, and dump positions. When a multipurpose type bucket is specified, an additional hydraulic control shall be supplied with clam close, hold, and open positions. If optional front-mounted or hydraulically operated rear-mounted equipment is specified (see 3.24.2), the necessary hydraulic connections and controls shall be furnished for convenient and easy operation. An automatic lift kickout and an automatic bucket positioner adjustable to desired loading angle shall be supplied unless otherwise specified.

3.16 Lighting. A full complement of adequate traveling lights shall be provided including minimum two front driving lights, two rear working lights, a stoplight and a taillight and instrument panel lighting. When specified (see 3.24.1), an amber flashing warning beacon light shall be mounted on the top of the cab with control switch marked and conveniently located inside the cab. When specified (see 3.24.1) front working lights with protective guards shall be provided.

3.17 Horn. Air or electric horn shall be provided on all loaders. Unless otherwise specified (see 3.24.1), a reverse signal alarm, audible type, shall be supplied conforming to 29 CFR 1926.602.

3.18 Instruments and gauges. The manufacturer's standard instruments and gauges shall be provided, including, but not limited to, ammeter or voltmeter, hour-meter, and gauges for engine oil pressure, fuel, engine coolant temperature (unless air-cooled engine is furnished), torque converter oil or hydraulic drive oil temperature, and air or hydraulic brake pressure for power assist portion of service brake system. A fuel dip stick is acceptable for a fuel gauge. When specified (see 3.24.1), an instrument panel guard that can be locked shall be furnished unless the loader is equipped with a lockable, enclosed cab. When specified, all gauges shall be of the mechanical type with lights (see 3.24.1).

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3.19 Security protection. When specified (see 3.24.1), a locking fuel tank cap, hydraulic oil tank cap, engine oil filler cap, engine oil dipstick, and radiator cap (unless air-cooled engine is furnished), shall be provided, unless protected by engine hood and easy access hood side doors or panels that can be locked.

3.20 Safety and comfort. Exposed parts that are subject to high operating temperatures shall be insulated, or guarded to prevent hazards to operating personnel. When specified (see 3.24.1), loader shall be equipped with fenders on all four tires to shield operator from debris thrown by tires. Moving parts which are a hazard to operating personnel shall be guarded or enclosed. The loader shall comply with 29 CFR 1926.602 (seatbelts, brakes, and horn), CFR 1926.604 (overhead covering), 29 CFR 1926,1000 (ROPS). When a ROPS with windshield is specified (see 3.24.1(r)), or when fully enclosed cab is specified (see 3.24.1(s)), it shall include glass (safety or equivalent) conforming to 29 CFR 1926.600 and rollover protection conforming to 29 CFR 1926.1000. Well-anchored seatbelts shall be furnished conforming to SAE J386, Type I.

3.21 Maintenance. A maintenance plate or polyester placard illustrating lubrication points and indicating servicing and maintenance instruction, intervals, and requirements shall be permanently affixed near the operator's compartment. Tools normally furnished with the loader by the manufacturer shall be provided. When specified (see 3.24.1), a toolbox with lock and keys shall be provided.

3.22 Painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt corrosion products, or any other interfering substances. As soon as practicable, after cleaning, and before any corrosion product or other coating interfering material can result, the surface shall be prepared or treated to insure that adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Unless otherwise specified (see 6.2) painting shall be in accordance with the manufacturer's standard painting procedure. The paint shall be free from runs, sags, orange peel, or other defects. Unless otherwise specified, loader and attachments shall be the same color.

3.23 Manufacturer's identification. The loader and allied equipment shall bear the manufacturer's name or trademark, model designation, and serial number on a plate securely affixed in a conspicuous place. The manufacturer's name or trademark may be cast integral, stamped, or otherwise permanently marked upon the equipment.

3.24 Optional accessories and allied equipment. When optional accessories and allied equipment are specified and furnished, such equipment shall be of current models so designed and constructed as to be readily adaptable without modification and shall operate efficiently with the vehicle so equipped.

3.24.1 Optional accessories. When specified in the invitation for bids, contract, or order (see 6.2), loaders shall be furnished with any of the following or other accessories as specified by the procuring agency:

(a) Block heater (see 3.7.2.2 for proper wattage starting requirements).
(b) Spark arrester, if engine is not turbocharged (see 3.7.2.7).
(c) Crankcase guard or integral designed crankcase guard (see 3.7.2.9).
(d) Power transferring differential in front or rear axle on articulated machines only. Not normally used on loaders over 2½ yd (see 3.10).
(e) Special tire treads and/or ply rating (see 3.12).
(f) Teeth (replaceble type) for general purpose bucket.
(g) Multipurpose bucket with replaceable type teeth instead of general purpose bucket.
(h) Spade nose rock bucket instead of general purpose bucket for size LW-3 and larger.
(i) Armrests, unless loader is equipped with bucket seat (see 3.15.1).
(j) OSHA approved reverse signal alarm, not required (see 3.17).
(k) Front working lights with protective guards, in addition to the lighting required under 3.16.
(l) Flashing warning beacon light (see 3.16).
(m) Mechanical type gauges with lights (see 3.18).
(n) Instrument panel guard, except when lockable cab is provided (see 3.18).

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(o) Locking type fuel tank, hydraulic oil tank, engine oil filler and radiator caps, when radiator is furnished (see 3.19), or lockable type hood, hood side doors enclosing caps.
(p) ROPS, with canopy top, windshield and powered windshield wiper(s), with windshield washer conforming to 29 CFR 1926.1000 and CFR 1926.600 (see 3.20).
(q) ROPS, cab, fully closed lockable type. The cab shall be the manufacturer's current standard production type enclosing the operator's seat and all controls. It shall be equipped with a minimum of one power-operated front windshield wiper and windshield washer. All glass shall conform to 29 CFR 1926.600 (see 3.20). When specified, personnel heater, air conditioner, defroster, defroster fan, and power-operated rear window wiper(s), shall be furnished. If cab is constructed with a split front windshield, an additional power-operated front windshield wiper shall be provided.
(r) Emergency steering in accordance with SAE J53 (see 3.13)
(s) Tow hooks, or eyes, front and/or rear, unless loader is equipped with rear mounted equipment.
(t) Fire extinguisher conforming to O-E-915, size 1-1/2 or 2-3/4, mounted in a position readily accessible to the operator.
(u) Toolbox with lock and keys (see 3.21).
(v) Clevis or pin type drawbar rear mounted, unless other rear mounted equipment will not permit mounting of drawbar.
(w) Sun canopy for shading operator. Not applicable with rollover protection.
(x) Fenders on all four tires (see 3.20).
(y) Warning horn that sounds when various temperatures or pressures reach critical levels (specify components).

3.24.2 Optional allied equipment. When specified in the invitation for bids, contract, or order (see 6.2), loaders shall be furnished with any of the following or other allied equipment as specified by the procuring agency. Allied equipment shall be of heavy-duty construction and be capable of handling the full power output of the specified loader under severe operating conditions. Equipment for attachment to, and use with loaders, shall be approved by the loader manufacturer for use with the loader model offered. Rear mounted equipment shall not be used to comply with minimum tipping load and minimum weight, unless the specifications for the rear mounted equipment as listed below specifically indicates that "this rear mounted equipment is to be permanently mounted."

(a) Backhoe for rear mounting on loaders size LW-3 and smaller. Specify type and width bucket required and minimum digging depth.
(b) Rear mounted ripper or scarifier. Specify number of teeth.
(c) Side dump bucket (specify size).
(d) Quick coupler system for loader attachments.
(e) Forklift (specify load capacity, tine length, 48 or 72 ins., hydraulic or manual tine separation; if forklift carriage oscillation required; if a backguard is required).
(f) Logging forks.
(g) Rear mounted winch.
(h) Rear mounted sliding coupler (for moving railroad cars).
(i) Snowplow (V, one-way or reversible).
(j) Snow blower.
(k) Tire chains.
(l) Feller buncher arrangement (specify type of guarding).

3.25 Special requirements.

3.25.1 Winterization. When specified (see 6.2), loaders shall be winterized to operate in -20°F., -40°F., or -65°F. temperatures.

3.25.2 Radio-interference suppression. When specified (see 6.2), the electrical system of the loader shall be radio-interference suppressed in the manner specified.

3.25.3 Fungus resistance. When specified (see 6.2), the loader electrical components shall be treated for fungus resistance.

3.25.4 Tiedown and/or lifting devices. When specified (see 6.2) loader shall be equipped with tiedowns and/or lifting devices in the manner specified in MIL-STD-209.

3.26 Workmanship. Workmanship shall conform to current best manufacturing practice followed for this type of equipment. Component parts and units shall be manufactured to definite standard dimensions with proper fits and clearances.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection. Each loader shall be given a general inspection for material, workmanship, and general compliance with the requirements of this specification.

4.3 Test. Compliance with table I shall be determined from the manufacturer's certified power curve, verified for diesel engines by regular production dynamometer tests with standard allowance for friction variations in new engines, and power deductions for fan, generator or alternator, and water pump if not on engine during test (see 3.7.1). Compliance with paragraph 3.4.1, Temperature conditions, shall be manufacturer's certification.

4.3.1 Test. When specified, paragraphs 4.3.2, 4.3.3, 4.3.4, 4.3.5, 4.3.6, 4.3.7, 4.3.8, 4.3.9, 4.3.10, and 4.3.11 shall be for military agencies only unless otherwise specified in this specification.

4.3.2 Lubrication. Lubrication means shall be provided for all moving parts requiring lubrication. Hydraulic lubrication fittings shall be in accordance with Specification SAE J534. Pressure lubrication fittings shall not be used where normal lubricating pressure may damage grease seals or other parts. Unless a winterized loader is specified, the loader shall be lubricated prior to delivery with lubricants designed for use in 0 to 32°F. temperature. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

4.3.3 Preliminary examination. The complete loader shall be examined and inspected to determine acceptability on the basis of conformance to contract requirements. Loader to be weighed for compliance with table I specification requirements.

4.3.4 Digging test. Excavate in undisturbed soil, or soil compacted to approximately the same state, provided it is as hard or harder than undisturbed soil and load into 5-ton dump trucks or frame work of comparable dimension for (2) continuous hours. During this period, a minimum of 30 full operating loads shall be handled each hour at normal working speed. Approximately 1/2 of this operation shall be conducted with loads clammed, from loose material, and bottom dumped. Check digging depth, dumping height and dumping reach (table II requirements). Periodic inspection during testing will be made to check for overheating and signs of breakage.

4.3.5 Stockpile loading test. A stockpile of loose soil, equal in volume to 10 times the volume of the capacity of the bucket, shall be loaded and transported a distance of at least 300 feet over uneven terrain. This operation shall not require more than 11 round trips per hour, leaving pickup area relatively clean. The operation shall be repeated for (2) consecutive hours without malfunction or overheating. The loader shall have sufficient power to crowd the bucket into a stockpile with sufficient force to fill the bucket to its rated size capacity in one pass.

4.3.6 Bulldozing test. The loader with bucket set for dozing cut of approximately 4-inches shall be employed to doze undisturbed soil or soil compacted to approximately natural state, provided it is as hard as or harder than undisturbed soil, into a stockpile approximately 15 times the bucket capacity in volume, and then spreading the same pile to a level surface. This operation shall not require more than 1 hour for completion. This operation shall be repeated for two (2) continuous hours.

4.3.7 Clamshell operation. Using the bucket as a clamshell, load loose soil or rock to rated bucket capacity from a stockpile to a 5-ton dump truck or framework of equivalent dimensions using the bottom dump feature of the bucket. A minimum of 50 passes shall be made in two (2) continuous hours. The bucket shall be used to pick up irregular shaped items such as brush, large tree trimmings, and large rocks by means of the edge grip and be capable of holding them firmly while traveling in carry position.

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4.3.8 Scraper operation. With the bucket partially opened and adjusted for a digging depth of 2 inches, the loader shall fill the bucket to its rated load capacity in undisturbed soil and deposit same on a stockpile or into a truck for 10 passes.

4.3.9 Tipping load test. The loader at standard operating weight shall be tilted back and positioned at the maximum forward reach in the raising cycle. The bucket shall be loaded until the rear wheels clear the surface to determine the tipping load.

4.3.10 Backhoe test. The backhoe attachment shall be assembled, adjusted and hydraulically operated to determine ease of installation and adequacy of design for the intended service and conformance to the specified requirements. The backhoe shall be operated in undisturbed soil for not less than 2 hours. Rear mounted equipment shall not be used to comply with minimum tipping load and minimum weight.

4.3.11 Test procedure. Testing in paragraphs 4.3.2, 4.3.3, 4.3.4, 4.3.5, 4.3.6, and 4.3.7 shall be conducted without the backhoe attached to the loader and with counterweight attached, if required. Failure of the backhoe to perform the tests specified shall be considered cause for rejection. Any equipment malfunction during the testing that requires repair prior to acceptance by the Government inspection shall also be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. Preservation and packaging shall be in acordance with MIL-T-3351, type II when specified (see 6.1).

5.1.2 Level C. Unless otherwise specified (see 6.1) the loaders shall be preseved and packaged for transport (mobile) in accordance with the supplier's standard practice, providing that this insures protection for the loader during shipment, and provides for safe delivery to their destination.

5.2 Packaging.

5.2.1 Levels A and B. Loaders preserved and packaged as specified in 5.1.1 shall be packed in accordance with MIL-T-3351, as specified for level A or B, type II, when specified.

5.2.2 Level C. Unless otherwise specified (see 6.1), the loaders shall be prepared for shipment (mobile) to insure carrier acceptance and safe delivery to destination in compliance with the Uniform Freight Classification rules or the Naitonal Motor Freight Classification rules depending upon mode of transportation utilized.

5.3 Marking.

5.3.1 Military agencies. In addition to any special markings required by the contract or order (see 6.1), the interior packages, shipping containers, semiboxed and mobile units shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Size loader required (see 1.2.1 and tables I, II, and III).
(c) Cold weather performance, must specify (see 3.4.1).
(d) Instructions for preparation for delivery if other than level C/C (see section 5).
(e) Manual publications (see 6.4).
(f) Illustrated literature (see 6.4.1).

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6.2 Optional ordering data. Unless the options in this specification are exercised, standard equipment shall be furnished. The following options, if required, should be specified.

(a) Painting, if different (see 3.22).
(b) Optional accessories, if required (see 3.24.1).
(c) Optional allied equipment, if required (see 3.24.2).
(d) Winterizing, if required (see 3.25.1 and 3.4.1).
(e) Tire hydroinflation, for additional traction only (see 3.4.3).
(f) Fungus resistance, if required (see 3.25.3).
(g) Radio-interference suppression, if required (see 3.25.2).
(h) Tiedowns and/or lifting devices, if required (see 3.25.4).
(i) Specify if manufacturer's field maintenance is required (see 6.5.3).
(j) Repair parts and service information, if required (see 6.5).
(k) Specify if Statement of Origin or Bill of Sale is required (see 6.6).

6.3 Warranty. When specified, the invitation for bids shall include provisions for a manufacturer's warranty with this specification similar to that specified in 6.3.1.

6.3.1 Warranty. The successful bidder shall be required to guarantee the loaders and parts thereof against defective material and workmanship for a period of 12 months or 2,000 operating hours, whichever occurs first, from date of acceptance of the equipment. However, if the contractor receives from any supplier or subcontractor additional warranty on the whole or any component of the loader, in the form of time or mileage, including any prorata arrangements, or the contractor general extends to his commercial customers a greater or extended warranty coverage, the Government shall receive corresponding warranty benefits. The guarantee shall include the furnishing without cost to the Government, f.o.b. manufacturer's nearest dealer or branch, or to the original destination if desired by the Government, of new parts and assemblies to replace any that prove to be defective within the warranty period. In addition, when the Government elects to have the work performed by the contractor, the cost of the labor involved in the replacement of the defective parts or assemblies shall be borne by the contractor. The contractor shall bear all costs of transporting the loader to his branch headquarters or the travel cost necessary to send qualified service people to the loader location to make repairs under warranty. On equipment used outside the 50 States of the United States and the District of Columbia, the guarantee shall include the furnishing of new parts or assemblies to replace any returned to the contractor by the Government which prove to be defective. The replacement parts or assemblies shall be delivered by the contractor to the port of embarkation in the United States designated by the Government. The contractor shall further guarantee that the loaders conform to all Federal laws and regulations governing noise, pollution, and other health and safety requirements which are required of the manufacturer of this type of equipment at the time of contract award. The contractor shall not be required to bear the cost of labor involved in correcting defects in equipment used outside the 50 States of the United States and the District of Columbia.

6.4 Manuals, operating and servicing. The successful bidder shall furnish one operator's manual, one shop manual, one lubricating chart, and one parts book with each loader, when specified (see 6.1).

6.4.1 Specifications and literature. When specified, the bidder shall furnish with his bid two sets of illustrated descriptive literature to assist the purchasing and using agencies in determining whether the loaders offered are adequate to perform the work specified (see 6.1).

6.5 Repair parts and service availability and bid qualification. When specified, service information and invitations for bid shall require data on repair parts, services, and qualification for bidder as included in 6.5.1 and 6.5.2.

6.5.1 Repair parts and service. As the continuous operation of the loader contemplated by this specification is of utmost importance, it is necessary that the successful bidder be in a position to render prompt service and to furnish replacement parts. Accordingly, bidders shall indicate the extent of their nearest dealer and branch office to the requisitioning activity where stocks of repair parts are maintained and can be secured within a reasonable time after ordering by part number from the parts book.

6.5.2 Qualification and proof by bidder. The bidder shall furnish proof that the manufacturer of the loader offered has been engaged in the production of wheel loaders for at least two years and further, that he is bidding on the latest current model loader, except as modified by this specification.

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6.5.3 Manufacturer's field maintenance. When specified (see 6.2), upon receipt of the equipment at the final destination, the contractor shall furnish to the Government at no extra cost the service of a fully qualified, trained mechanic or serviceman to supervise and inspect the proper assembly and operation to the equipment. It is required that he completely adjust the loader, install all components, lubricate the equipment with the grade of lubricant recommended by the manufacturer, charge battery, and make the unit ready for continuous operation. The cooling system shall be filled to the proper operating level with manufacturer's standard, or if necessary due to weather conditions at the time of delivery and altitude of delivery point, with suitable noncorrosive, nonevaporating solution with a sufficiently low freezing point to permit normal operation without danger of freezing. In addition, the Government's operator shall be instructed in the proper operation and maintenance of the machine. All of the material required for the foregoing operations shall be furnished by the contractor.

The contractor shall also furnish a minimum of one free followup service call to be made by a competent, fully qualified trained mechanic or serviceman. This service call shall be made at such time, either within 45 days after complete delivery of the equipment or after 30 days usage, as may be designated after such delivery by the contracting officer or his duly authorized representative.

6.6 Statement of Origin or Bill of Sale. When specified, manufacturer's Statement of Origin or Bill of Sale showing the applicable Delivery Order Number is required for each vehicle procured under this specification. Unless othewise specified, such documents shall be forwarded to the consignee.

MILITARY INTERESTS:

Review Activities:

Army - ME
Navy - YD
Air Force - 85

CIVIL AGENCY COORDINATING ACTIVITIES:

USDA - AFS
DOT - FHW, MVP

PREPARING ACTIVITY

GSA

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, *fold, staple, and mail.* to GSA-OPP-YAE, CMBg. #4, Rm. 420, Wash., DC 20406

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

KKK-L-1542C, Loaders, Scoop Type, Four-Wheel Drive, Pneumatic-Tired.

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

GSA DC 69-9761

FORM GSA FEB 69 2200

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